Graph-Based Spatiotemporal Flood Arrival Prediction for Pinpointed Risk Classification

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Abstract

Flood Mortality is disproportionately high in under-resourced communities, specifically in communities that have a lack of, or have unreliable warning systems. Current systems focus on absolute stage thresholds at discrete gauge sites that have to depend upon spatiotemporal propagation dynamics critical to flash flood modelling and prediction. Project AEGIS (Adaptive Environmental Guardian & Information System) implements a graph-theoretic, attention-enhanced framework for predicting time-to-flood arrival at arbitrary points along a terrain. The approach integrates topological hydrodynamic propagation models parameterized by DEMs, velocity/acceleration sensitive predictive inference layers, and transfer learning (LSTM attention) trained on hydrological datasets. The system outputs node-wise temporal classifications and optimized evacuation vectors. Converse to the gauge based approaches prevalent, AEGIS follows a fallback system that preserves key functions under infrastructural degradation, scales to data-sparse regions, and uses ROC dynamics for high velocity flooding.

Introduction

Global flash flooding accounts for a minimum of 40% of flood induced fatalities [1]. The majority of these fatalities occur in non-urban, sensor sparse regions. Specifically, the Kerr County Texas flash flooding of July 4, 2025 exemplifies these systemic vulnerabilities.

Primarily we address:

- loss of cellular/wifi and power networks,
- county level warnings that provide no individual risk calculation,
- absence of advanced sensor networks, and

- exponential escalation of flooding.

Current operational models (for example: HEC-RAS, NOAA AHPS) are modeled upon and suit networks with a dense system of gauges and deterministic stage-discharge relationships, meaning generalization to locations with low-infrastructure budgeting is infeasible. Furthermore, models currently in use provide only height or inundation extents, often leaving out time-to-arrival forecasts, at terrain resolved scales, especially under non-linear acceleration dynamics.

System Architecture (3 parts)

Representation of Data

(*Topographical here refers to graph-theoretic connectivity of hydrological landscapes, not abstract algebra*)

We encode water propagation as nodes and edges capturing terrain dependent hydraulic connectivity where nodes V represent discrete terrain elements for (e.g: 30m cells from USGS NED DEM3) and edges E are directed links representing potential flow paths, derived via D8/D gradient-based flow accumulation algorithms [2].

Each node has the following attributes Elevation, Slope S_{ij} (gradient between adjacent nodes), soil infiltration capacity ϕ_i, and distance to nearest channel d_i.

Each edge has the following attributes weights (functions of local slope roughness, and hydraulic connectivity).

Resulting graph $G = (V, E)$ provides a dynamic topological basis that evolves with river stage, precipitation, and soil saturation. The graph's effective hydraulic head differences dictate flow propagation paths.

Inference Layer

Flood Arrival Times are predicted by making utilization of sequence-to-sequence LSTM networks with attention. This enables successful captures of time dependencies and prioritizes drivers upstream.

We input river stage time series $h(t)$ from USGS NWIS, river velocity $\frac{dh}{dt}$, precipitation intensity via NOAA NEXRAD radar, soil saturation index $S(t)$ via USDA SSURGO + changes in via local modelling.

And receive the following: Node-specific flood arrival probability distribution $p(i \mid X)$ and thus $\hat{i}$.

Why attention? Attention mechanism allows the network to dynamically weight upstream contributions to downstream hazards.

Velocity–Acceleration Hazards (completed)

We explicitly use rate-of-change to detect explosive-onset flash behavior before absolute-stage thresholds are crossed.

Definitions (per node i): instantaneous stage $h_i(t)$; velocity $v_i(t) = \frac{dh_i(t)}{dt}$; acceleration $a_i(t) = \frac{d^2h_i(t)}{dt^2}$.

Trigger thresholds:

$$v_i(t) > v_c \quad \text{(rapid rise)}, \qquad a_i(t) > a_c \quad \text{(onset of exponential escalation)}.$$

Thresholds (v_c, a_c) are calibrated per basin using historical events.

Kinetic hazard score: we combine dynamics and proximity without requiring the user to know where the river is:

$$H_i(t) = \gamma_1 \, v_i^2(t) \; + \; \gamma_2 \, a_i(t) \; + \; \gamma_3 \, \frac{v_i(t)}{\max(d_i, \varepsilon)},$$

with small ε to avoid division by zero. This converts "how fast water is rising upstream" into a localized risk signal that can propagate on the terrain graph to any GPS point downstream.

Explicit Propagation Math (no abstraction)

We turn upstream triggers into downstream time-to-arrival at every terrain cell.

Edge geometry: for an edge $i \to j$: center-to-center distance L_{ij}; slope $S_{ij} = \frac{e_i - e_j}{L_{ij}}$ (set $S_{ij} = 0$ if $e_j \geq e_i$); composite roughness $n_{ij} = \frac{r_i + r_j}{2}$.

Local velocity (Manning-style for gridded overland flow):

$$c_{ij} = \frac{1}{n_{ij}} \, S_{ij}^{1/2} \, R_{ij}^{2/3}, \quad R_{ij} \approx \frac{h_i \, L_{ij}}{L_{ij} + 2h_i}.$$

Edge travel time:

$$\tau_{ij} = \frac{L_{ij}}{\max(c_{ij}, c_{\min})}.$$

Arrival time field (wavefront update):

$$T_j = \min_{i \in U(j)} \big(T_i + \tau_{ij}\big),$$

with $U(j)$ the set of upstream contributors of j. Initialize T_i at triggered nodes (those exceeding v_c or a_c) using the current clock time; propagate with Dijkstra-like updates over directed downslope edges (D8/D∞).

Coupling dynamics to propagation: when $a_i(t)$ spikes, we shrink τ_{ij} by a bounded factor to reflect surge amplification:

$$\tau_{ij}^{\text{eff}} = \frac{\tau_{ij}}{1 + \kappa \cdot \sigma(a_i(t))}, \quad \sigma(x) = \frac{1}{1 + e^{-x}},$$

κ tuned by event replay. This keeps physics explicit (still downhill, still slope-controlled) while translating acceleration into earlier arrivals.

Risk Classification (pinpointed, user-facing)

We do not assume users know river proximity. We map their GPS to the nearest terrain node and use its predicted lead time and kinetic hazard.

$$\text{Lead time } \hat{T}_i \quad \text{from propagation/LSTM}, \quad \text{risk } R_i = \Pr(T_i < t_c) \cdot H_i(t).$$

Zones:

Red: $\hat{T}_i < 15$ min or $R_i \geq 0.75$; Yellow: $15 \leq \hat{T}_i < 60$ min or $0.35 \leq R_i < 0.75$; Blue: $\hat{T}_i \geq 60$ min and R

Evacuation Vector Optimization (explicit)

We output a minimum-cost path to safe elevation in the user's reachable graph neighborhood.

Raster-to-graph travel model: for movement edge $p \to q$:

$$\text{move_time}_{pq} = \frac{L_{pq}}{\max(u_{\min},\, u_{\text{base}} \cdot g(\theta_{pq}))},$$

where u_{base} is nominal walking/driving speed, θ_{pq} is slope angle, and $g(\theta)$ is a monotone decreasing function (e.g., $g(\theta) = \cos^{+}\theta$) with a cutoff on unsafe gradients.

Safety potential: penalize paths into predicted inundation during the travel horizon:

$$\text{cost}_{pq} = \text{move_time}_{pq} \; + \; \lambda_{\text{flood}} \cdot \mathbf{1}\{T_q < t_{\text{ETA}}(q)\}.$$

Shortest path (A^* or Dijkstra) from user node to any node with $e \geq e_{\text{safe}}$ and $\Pr(T > t_{\text{ETA}})$ high.

Data Sources (explicit)

- **Topography:** USGS NED DEM (30 m).
- **Hydrology:** USGS NWIS stage/flow archives.
- **Meteorology:** NOAA NEXRAD radar precipitation.
- **Soils/Landcover:** USDA SSURGO; NLCD 2019 roughness.
- **Validation footprints:** FEMA flood maps; Dartmouth Flood Observatory.

Training and Validation (what we actually do)

Foundation basins (data-rich) → transfer to sparse: Train on Houston (Buffalo Bayou), Austin (Colorado River), then fine-tune to sparse basins (e.g., Kerr County).

Supervision signals:

$$\begin{aligned} &\text{(1) Arrival regression: } \hat{T}_i \text{ vs observed arrival time} \quad \Rightarrow \text{ RMSE} \\ &\text{(2) Zone classification: } \{\text{Red, Yellow, Blue}\} \; \Rightarrow \; \text{F1/macro-F1} \\ &\text{(3) Consistency: } \hat{T}_j \geq \hat{T}_i + \tau_{ij} \text{ (soft constraint)} \end{aligned}$$

Input tensors (per node):

$$x_i(t) = [e_i,\ \phi_i,\ d_i,\ r_i,\ S_{ij},\ h_i(t),\ v_i(t),\ a_i(t),\ P_i(t),\ s_i(t)].$$

Static terms are concatenated to every time step. Sequence length T chosen per basin timescale.

Attention bias (hydro-aware):

$$\alpha_{ij} = \frac{\exp(Q_i^\top K_j + \beta_1 S_{ij} + \beta_2 e^{-d_i/\lambda})}{\sum_{k \in U(j)} \exp(Q_k^\top K_j + \beta_1 S_{kj} + \beta_2 e^{-d_k/\lambda})}.$$

Deployment and Fallback

Normal conditions: full stack (NEXRAD + NWIS + DEM + SSURGO/NLCD).

Degraded conditions (gauge/network loss): keep propagation alive using:

$$\frac{dh_i}{dt} = P_i(t) - \phi_i - \sum_j c_{ij} \, [h_i - h_j]^+,$$

and update T via graph travel times; push zones and evacuation vectors via local mesh (LoRa/siren beacons). No user river knowledge required.

Limitations (no hand-waving)

- Overland flow with grid-scale Manning is an approximation; channel hydraulics during bridge constrictions need explicit handling (culvert weirs).
- DEM resolution bounds accuracy; LiDAR improves S_{ij} and flow routing.
- Thresholds (v_c, a_c) are basin-specific; we calibrate by replay.
- Human movement speeds degrade under panic/obstacles; we include conservative buffers in path costs.

Conclusion

AEGIS converts upstream rate-of-change into explicit, graph-propagated arrival times and risk at every terrain cell. The user only needs a GPS point; the system internalizes the hydrology. Outputs: (1) a lead-time map (Red/Yellow/Blue), (2) a kinetic hazard scalar, and (3) an evacuation vector that avoids predicted inundation within the travel horizon. Works under degraded infrastructure, trains on rich basins, and transfers to sparse ones without breaking physics.

References

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